Singapore School Kinderland

Continual Assessment 1

2026 - 2027

Secondary 2

Computer Science

Part 1

Date: _____ Name: _____

Instructions:

- Start answering the questions on time.
- Follow all instructions carefully.
- Answer all questions.
- Submit your paper within the given time.

Total marks	**35**
Part 1	**30**
CW	**5**
Marks Obtained	

Section A: Multiple Choice Questions (MCQ) (6x1=6 marks)

For each question 1-6, four options are given. Circle the correct option i, ii, iii or iv.

1. Which digits are used in the Binary (Base-2) number system?

i. 0 and 1 only **ii.** 0 to 9

iii. 1 and 2 only **iv.** 0 to 9 and A to F

2. What is the Denary (Base-10) value of the 4-bit binary number 1010?

i. 8 **ii.** 10

iii. 12 **iv.** 15

3. In the Hexadecimal (Base-16) system, which Denary number is represented by the letter F?

i. 10 **ii.** 12

iii. 14 **iv.** 15

4. How many binary bits are grouped together to represent ONE Hexadecimal digit?

i. 2 bits **ii.** 4 bits (1 nibble)

iii. 8 bits **iv.** 16 bits

5. In Binary Addition, what is the result of adding 1 + 1?

i. 2 (Carry 0) **ii.** 0 (Carry 0)

iii. 0 with Carry 1 (10 in binary) **iv.** 1 with Carry 1 (11 in binary)

6. What is the largest Denary number that can be stored in an 8-bit byte (1111 1111)?

i. 128 **ii.** 255

iii. 256 **iv.** 512

Section B: Binary and Denary Conversions (4x2=8 marks)

7. Use the 8-bit place values (128, 64, 32, 16, 8, 4, 2, 1) to complete the conversions below:

No.	Conversion Task & Working Space	Final Answer
i.	Convert 8-bit Binary **0010 1101** to Denary (Base-10): Working: _____	
ii.	Convert 8-bit Binary **1001 0110** to Denary (Base-10): Working: _____	
iii.	Convert Denary **75** into an 8-bit Binary number: Working: _____	
iv.	Convert Denary **196** into an 8-bit Binary number: Working: _____	

Section C: Binary and Hexadecimal Conversions (4x2=8 marks)

8. Convert between Binary and Hexadecimal using 4-bit groups / nibbles (8, 4, 2, 1):

No.	Conversion Task & Working Space	Final Answer
i.	Convert 8-bit Binary **1010 0101** to Hexadecimal: Working: _____	
ii.	Convert 8-bit Binary **1111 0011** to Hexadecimal: Working: _____	
iii.	Convert Hexadecimal **4C** into an 8-bit Binary number: Working: _____	
iv.	Convert Hexadecimal **9E** into an 8-bit Binary number: Working: _____	

Section D: Binary Addition (8-Bit Numbers) (4x2=8 marks)

9. Add the following 8-bit binary numbers. Show your Carry row and Final 8-bit Sum clearly:

Binary Addition Rules: 0 + 0 = 0 | 0 + 1 = 1 | 1 + 0 = 1 | **1 + 1 = 0 (Carry 1)** | **1 + 1 + 1 = 1 (Carry 1)**

No.	8-Bit Binary Addition Problem	Carry & Final Sum
i.	0 0 1 0 0 1 0 1 + 0 0 0 1 0 0 1 0	Sum: _____ Carry: _____
ii.	0 1 0 0 1 0 1 1 + 0 0 1 0 0 1 1 0	Sum: _____ Carry: _____
iii.	0 1 0 1 1 1 0 0 + 0 0 1 1 1 0 1 0	Sum: _____ Carry: _____
iv.	0 1 1 0 1 1 0 1 + 0 1 0 1 0 1 1 1	Sum: _____ Carry: _____